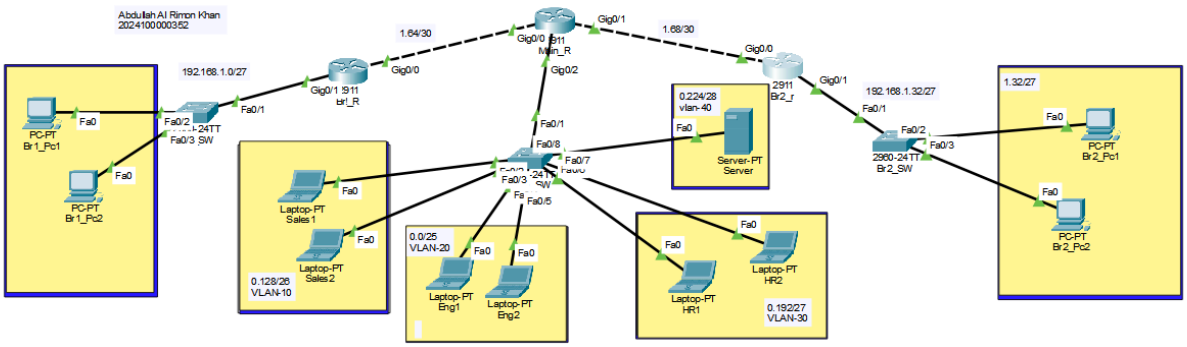


Main_SW

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config)#interface range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface range fa0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#interface range fa0/6-7
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#interface fa0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write memory
Building configuration...
[OK]
Switch#show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1, Fa0/9, Fa0/10, Fa0/11
                                                 Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                 Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                 Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                 Fa0/24, Gig0/1, Gig0/2
10   sales                             active    Fa0/2, Fa0/3
20   engineering                       active    Fa0/4, Fa0/5
30   HR                                active    Fa0/6, Fa0/7
40   server                            active    Fa0/8
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
```

```
[OK]
Main_R#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.65    YES manual up                    down
GigabitEthernet0/1     192.168.1.69    YES manual up                    down
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/2.10  192.168.0.129   YES manual up                    up
GigabitEthernet0/2.20  192.168.0.1     YES manual up                    up
GigabitEthernet0/2.30  192.168.0.193   YES manual up                    up
GigabitEthernet0/2.40  192.168.0.225   YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Main_R#
```

Br!_R

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#inter gi0/0
Router(config-if)#ip add 192.168.1.66 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up

Router(config-if)#inter gi0/1
Router(config-if)#ip add 192.168.1.1 255.255.255.224
Router(config-if)#ip helper
Router(config-if)#ip helper-address 192.168.0.226
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

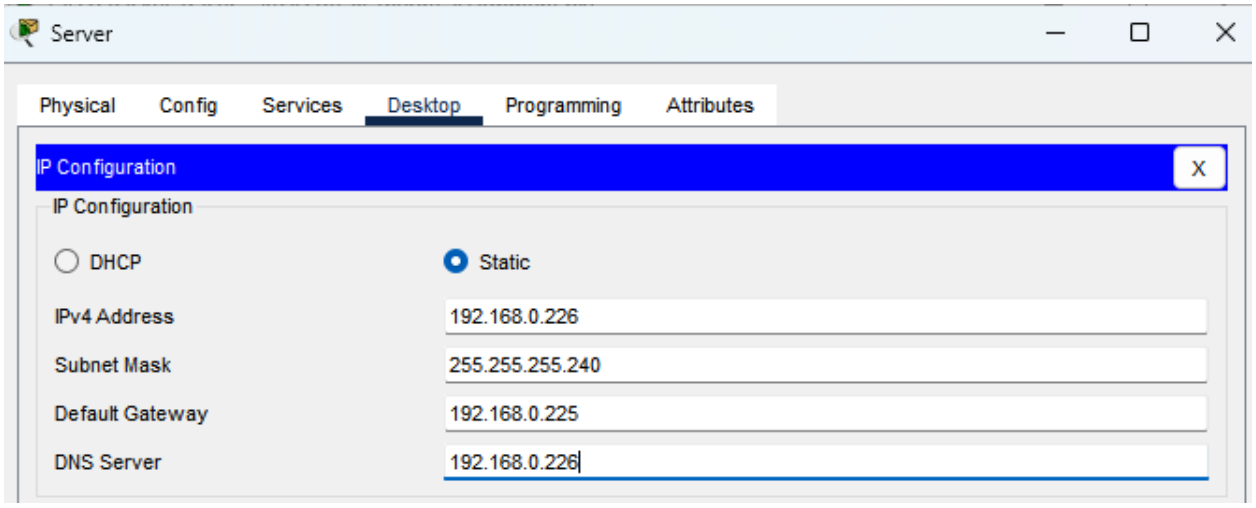
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
up

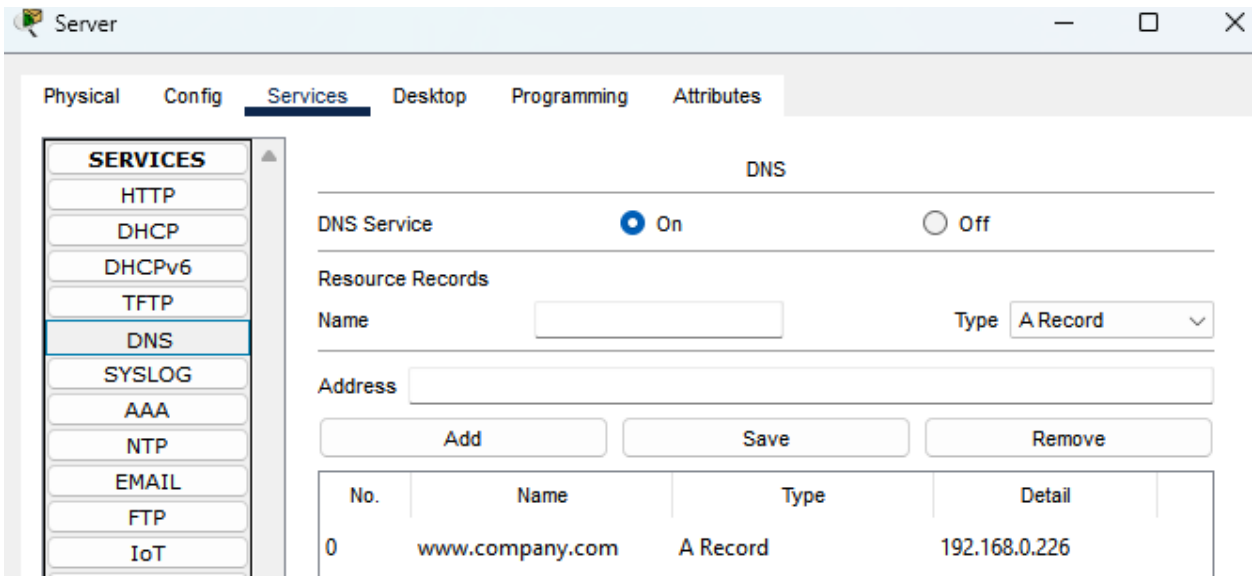
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.65
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

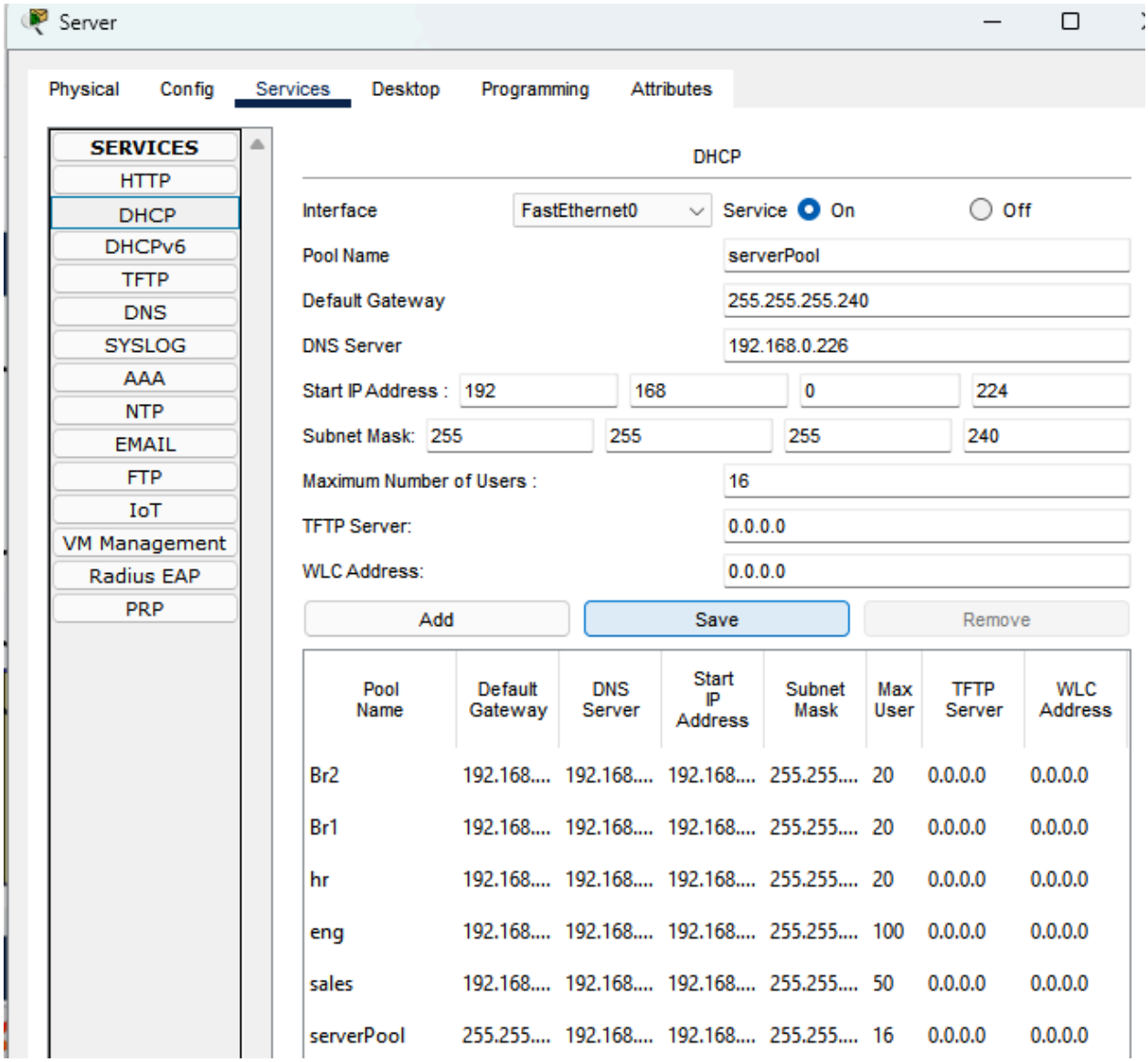
Router#write memory
Building configuration...
[OK]
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.66    YES manual up                    up
GigabitEthernet0/1     192.168.1.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unass           unset  administratively down down
```

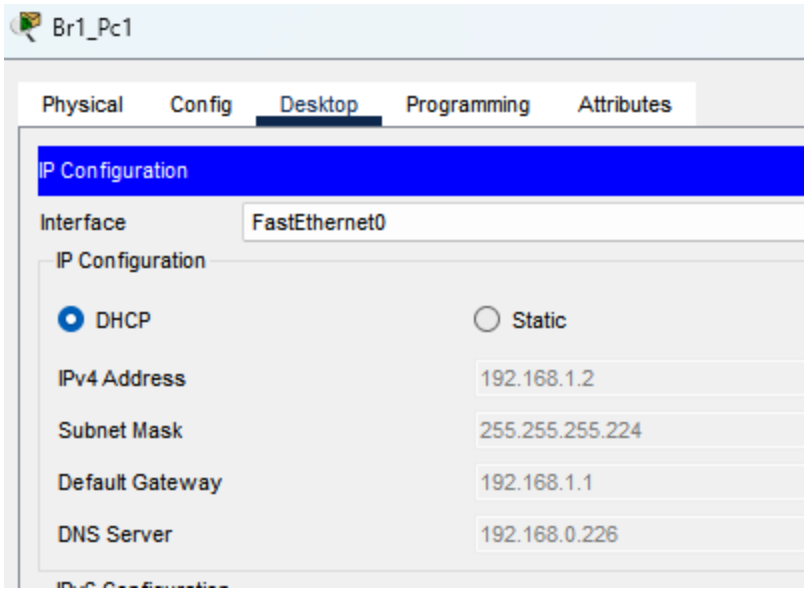
Microsoft Edge

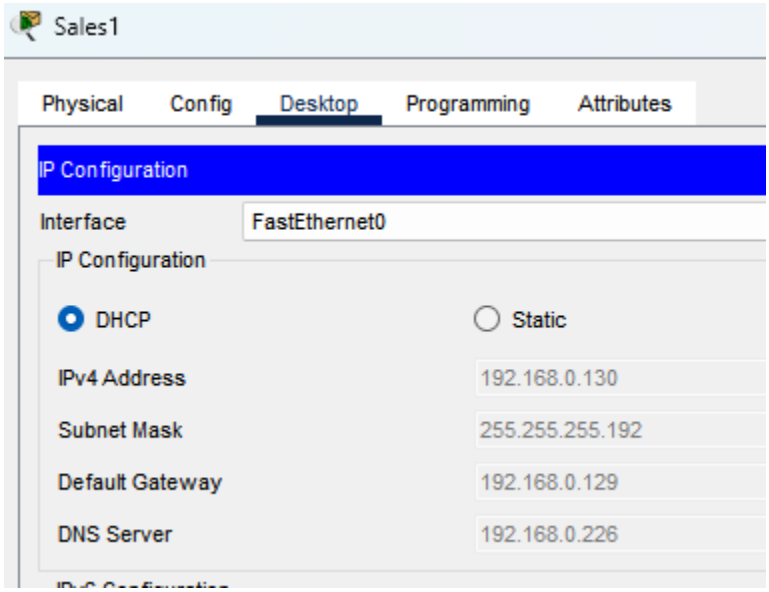
```
Br2_r#show ip inter
Br2_r#show ip interface b
Br2_r#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.70    YES manual up                    up
GigabitEthernet0/1     192.168.1.33    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Br2_r#
```

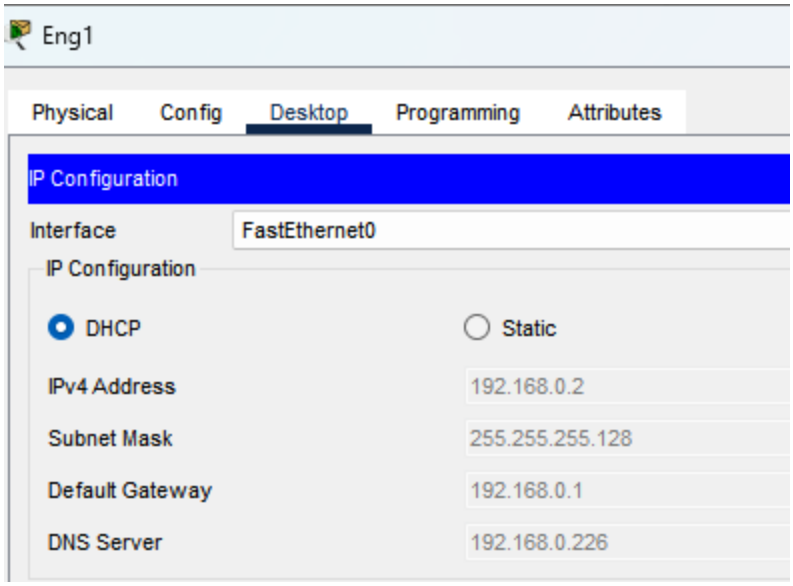


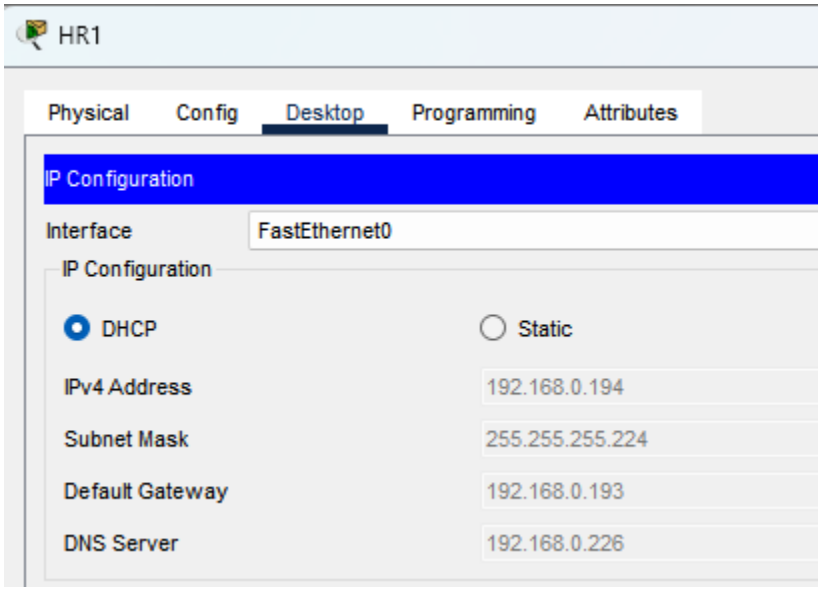


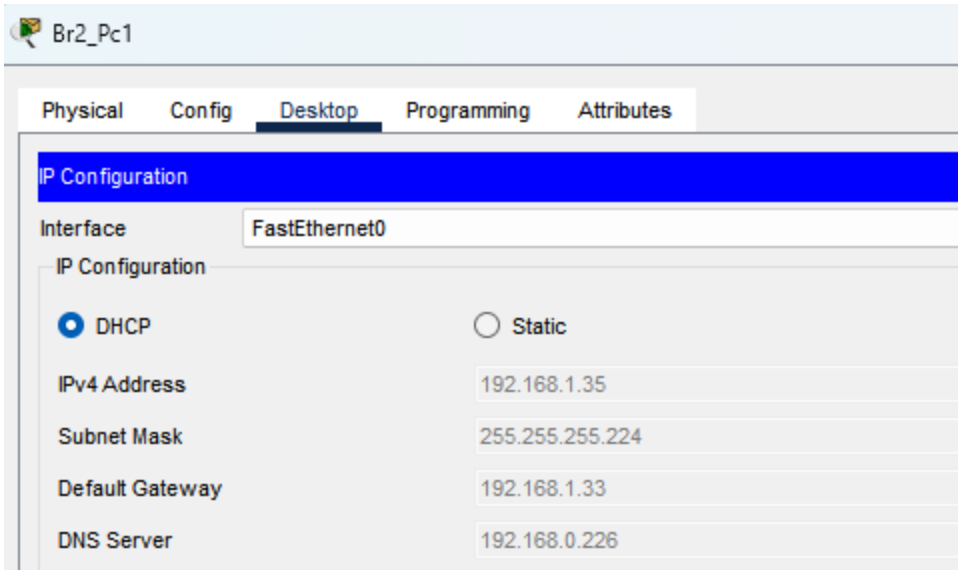

Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
255.255.255.240
DNS Server
192.168.0.226
Start IP Address :
192
168
0
224
Subnet Mask:
255
255
255
240
Maximum Number of Users :
16
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
Br2 192.168.... 192.168.... 192.168.... 255.255.... 20 0.0.0.0 0.0.0.0
Br1 192.168.... 192.168.... 192.168.... 255.255.... 20 0.0.0.0 0.0.0.0
hr 192.168.... 192.168.... 192.168.... 255.255.... 20 0.0.0.0 0.0.0.0
eng 192.168.... 192.168.... 192.168.... 255.255.... 100 0.0.0.0 0.0.0.0
sales 192.168.... 192.168.... 192.168.... 255.255.... 50 0.0.0.0 0.0.0.0
serverPool 255.255.... 192.168.... 192.168.... 255.255.... 16 0.0.0.0 0.0.0.0

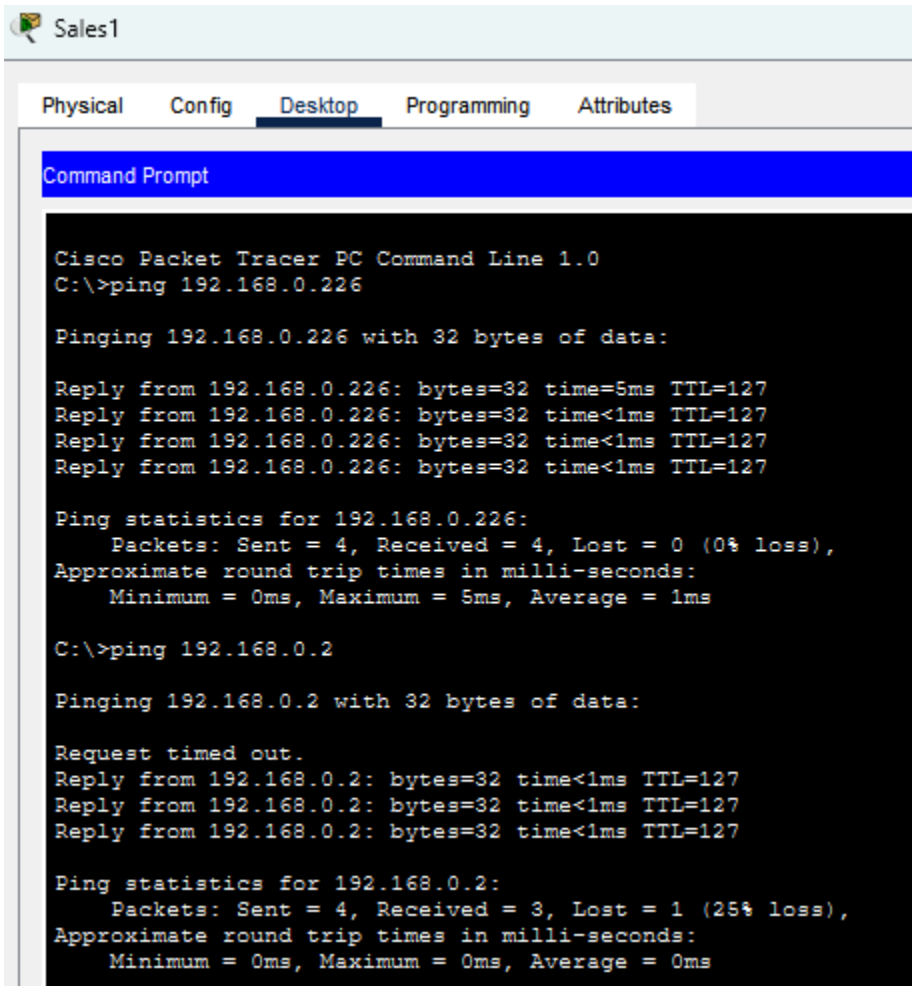

Br1_Pc1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.224
Default Gateway
192.168.1.1
DNS Server
192.168.0.226

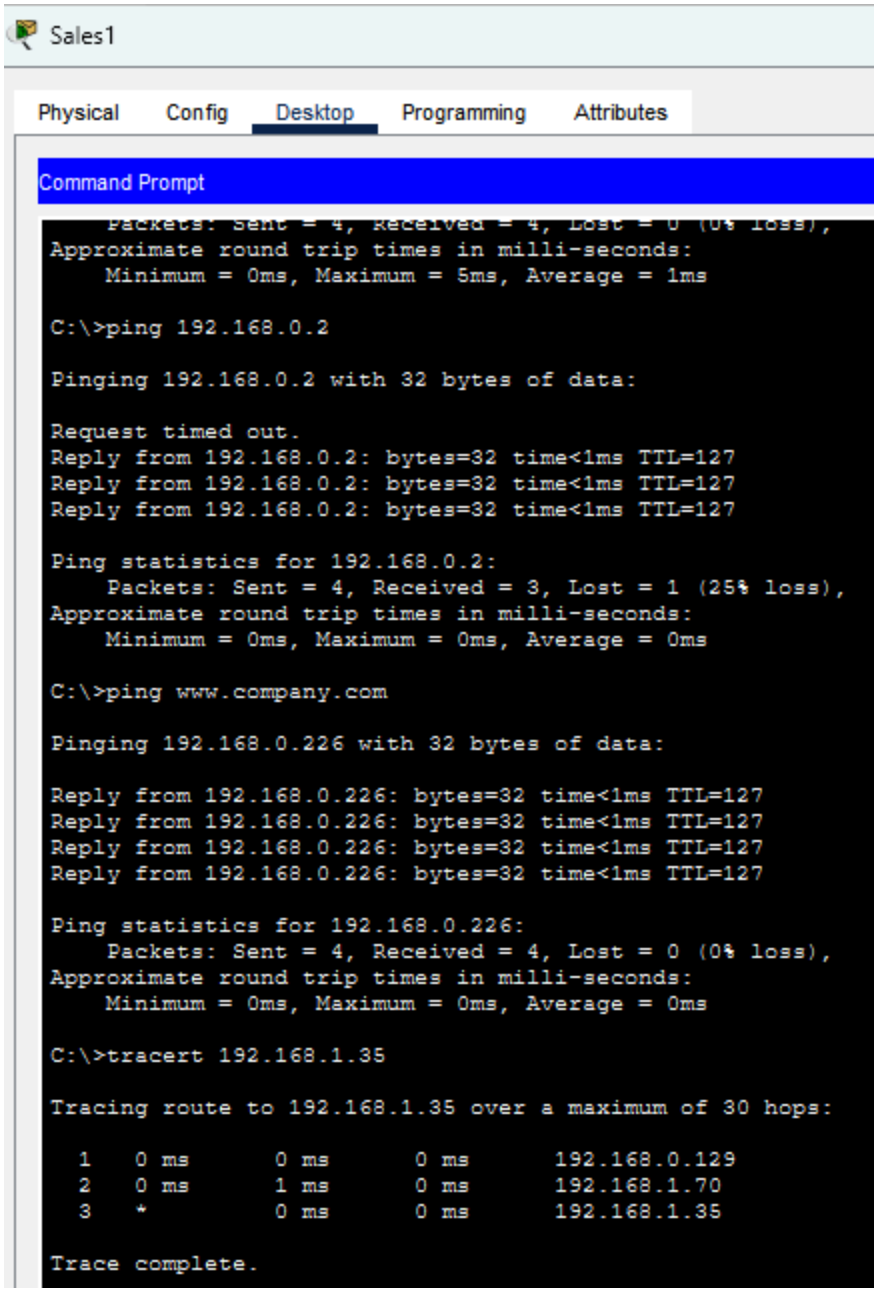

Sales1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.0.130
Subnet Mask
255.255.255.192
Default Gateway
192.168.0.129
DNS Server
192.168.0.226


Eng1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.0.2
Subnet Mask
255.255.255.128
Default Gateway
192.168.0.1
DNS Server
192.168.0.226


HR1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.0.194
Subnet Mask
255.255.255.224
Default Gateway
192.168.0.193
DNS Server
192.168.0.226


Br2_Pc1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.35
Subnet Mask
255.255.255.224
Default Gateway
192.168.1.33
DNS Server
192.168.0.226


Sales1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.226
Pinging 192.168.0.226 with 32 bytes of data:
Reply from 192.168.0.226: bytes=32 time=5ms TTL=127
Reply from 192.168.0.226: bytes=32 time<1ms TTL=127
Reply from 192.168.0.226: bytes=32 time<1ms TTL=127
Reply from 192.168.0.226: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.0.226:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>ping 192.168.0.2
Pinging 192.168.0.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.0.2: bytes=32 time<1ms TTL=127
Reply from 192.168.0.2: bytes=32 time<1ms TTL=127
Reply from 192.168.0.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.0.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms


Sales1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>ping 192.168.0.2
Pinging 192.168.0.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.0.2: bytes=32 time<1ms TTL=127
Reply from 192.168.0.2: bytes=32 time<1ms TTL=127
Reply from 192.168.0.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.0.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping www.company.com
Pinging 192.168.0.226 with 32 bytes of data:
Reply from 192.168.0.226: bytes=32 time<1ms TTL=127
Reply from 192.168.0.226: bytes=32 time<1ms TTL=127
Reply from 192.168.0.226: bytes=32 time<1ms TTL=127
Reply from 192.168.0.226: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.0.226:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>tracert 192.168.1.35
Tracing route to 192.168.1.35 over a maximum of 30 hops:
1 0 ms 0 ms 0 ms 192.168.0.129
2 0 ms 1 ms 0 ms 192.168.1.70
3 * 0 ms 0 ms 192.168.1.35
Trace complete.

PDU Information at Device: Sales1

OSI Model | Outbound PDU Details

At Device: Sales1
Source: Sales1
Destination: Eng2

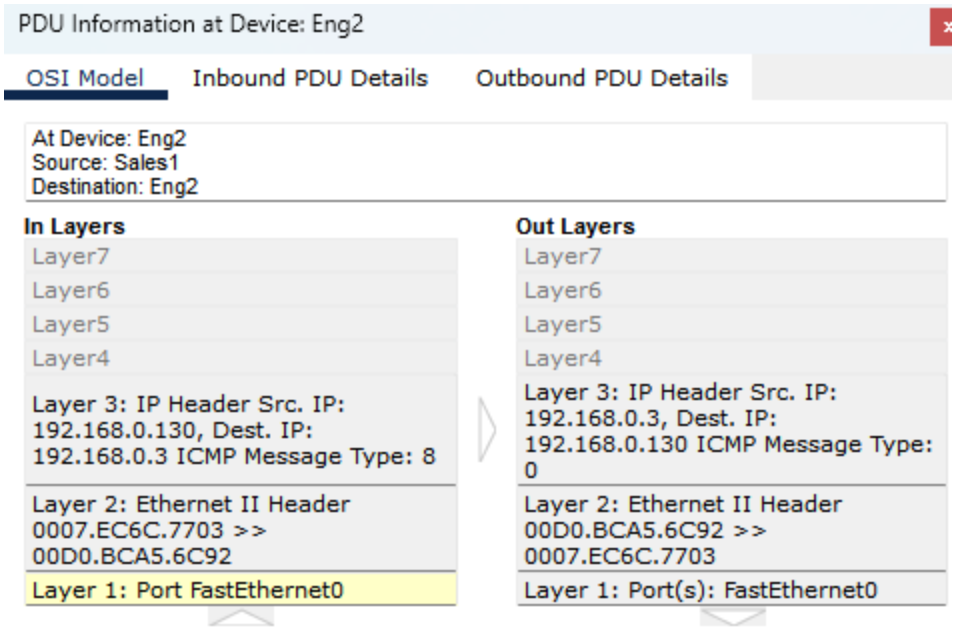
In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer 3: IP Header Src. IP: 192.168.0.130, Dest. IP: 192.168.0.3 ICMP Message Type: 8
Layer2	Layer 2: Ethernet II Header 0030.A3E8.C9B2 >> 0007.EC6C.7703
Layer1	Layer 1: Port(s): FastEthernet0

PDU Information at Device: Main_R

OSI Model | Inbound PDU Details | Outbound PDU Details

At Device: Main_R
Source: Sales1
Destination: Eng2

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer 3: IP Header Src. IP: 192.168.0.130, Dest. IP: 192.168.0.3 ICMP Message Type: 8	Layer 3: IP Header Src. IP: 192.168.0.130, Dest. IP: 192.168.0.3 ICMP Message Type: 8
Layer 2: Dot1q Header 0030.A3E8.C9B2 >> 0007.EC6C.7703	Layer 2: Dot1q Header 0007.EC6C.7703 >> 00D0.BCA5.6C92
Layer 1: Port GigabitEthernet0/2	Layer 1: Port(s): GigabitEthernet0/2


PDU Information at Device: Eng2
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: Eng2
Source: Sales1
Destination: Eng2
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.0.130, Dest. IP: 192.168.0.3 ICMP Message Type: 8
Layer 2: Ethernet II Header 0007.EC6C.7703 >> 00D0.BCA5.6C92
Layer 1: Port FastEthernet0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.0.3, Dest. IP: 192.168.0.130 ICMP Message Type: 0
Layer 2: Ethernet II Header 00D0.BCA5.6C92 >> 0007.EC6C.7703
Layer 1: Port(s): FastEthernet0

PDU Information at Device: Main_R

OSI Model | Inbound PDU Details | Outbound PDU Details

At Device: Main_R
Source: Sales1
Destination: Eng2

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer 3: IP Header Src. IP: 192.168.0.3, Dest. IP: 192.168.0.130 ICMP Message Type: 0	Layer 3: IP Header Src. IP: 192.168.0.3, Dest. IP: 192.168.0.130 ICMP Message Type: 0
Layer 2: Dot1q Header 00D0.BCA5.6C92 >> 0007.EC6C.7703	Layer 2: Dot1q Header 0007.EC6C.7703 >> 0030.A3E8.C9B2
Layer 1: Port GigabitEthernet0/2	Layer 1: Port(s): GigabitEthernet0/2

PDU Information at Device: Sales1

OSI Model | Inbound PDU Details

At Device: Sales1
Source: Sales1
Destination: Eng2

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer 3: IP Header Src. IP: 192.168.0.3, Dest. IP: 192.168.0.130 ICMP Message Type: 0	Layer3
Layer 2: Ethernet II Header 0007.EC6C.7703 >> 0030.A3E8.C9B2	Layer2
Layer 1: Port FastEthernet0	Layer1